SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

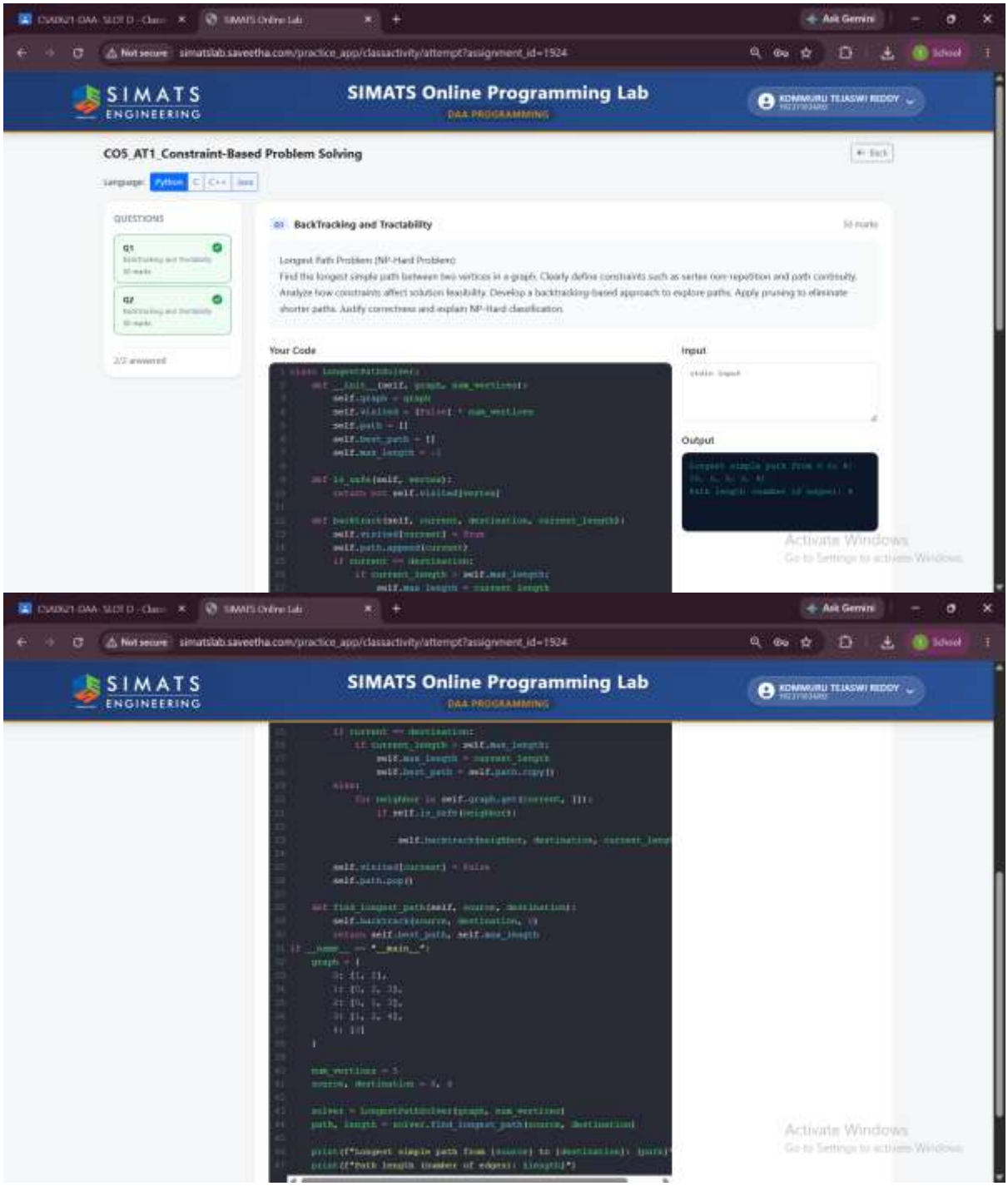
Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO5– Naturalization (BL5,BL6)	**Assessment:**	Assessment Tool 1 – Constraint Based Problem Solving
Weightage:	30% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO5 Statement:	*Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.*		
Student Name:	K Tejaswi Reddy	**Reg. No.:**	192371036
Section / Batch:	CS-BIOSCI	**Date:**	29-08-2026

Instructions to Students

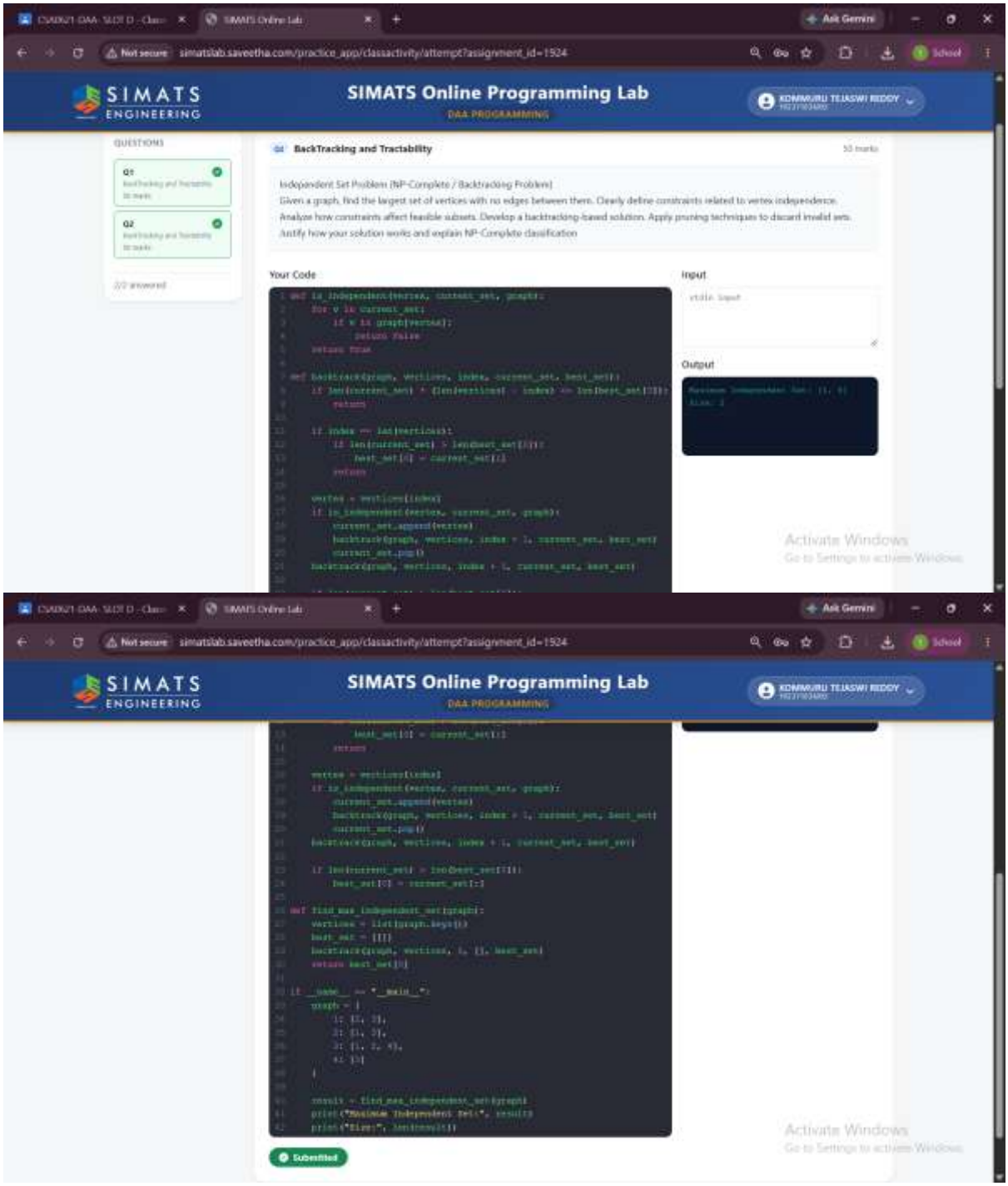
- • Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
  • Apply backtracking techniques systematically for combinatorial problem solving.
  • Clearly classify problems under P, NP, NP-Hard, and NP-Complete categories wherever required.
  • Provide proper justification, state-space representation, and complexity analysis for all solutions.
  • Neat presentation and logical explanation are mandatory.

Question Type	Focus Area	Questions	Marks
Constraint Based Problem Solving	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NP-Complete.	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – CONSTRAINT BASED PROBLEM SOLVING


SIMATS ENGINEERING
SIMATS Online Programming Lab
DAA PROGRAMMING
CO5_AT1_Constraint-Based Problem Solving
Back
Language: Python C C++ Java
QUESTIONS
Q1
Q2
2/2 answered
Q1 BackTracking and Tractability
10 marks
Longest Path Problem (NP-Hard Problem)
Find the longest simple path between two vertices in a graph. Clearly define constraints such as series non-repetition and path continuity. Analyze how constraints affect solution feasibility. Develop a backtracking-based approach to explore paths. Apply pruning to eliminate shorter paths. Justify correctness and explain NP-Hard classification.
Your Code
Input
Output
Activate Windows
Go to Settings to activate Windows.

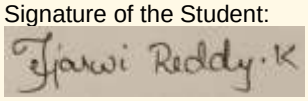
SIMATS ENGINEERING
SIMATS Online Programming Lab
DAA PROGRAMMING
Activate Windows
Go to Settings to activate Windows.


SIMATS ENGINEERING
SIMATS Online Programming Lab
DAA PROGRAMMING
KOMMURU TEJASWI REDDY
QUESTIONS
Q1
Q2
Q4 BackTracking and Tractability
Independent Set Problem (NP-Complete / Backtracking Problem)
Given a graph, find the largest set of vertices with no edges between them. Clearly define constraints related to vertex independence.
Analyze how constraints affect feasible subsets. Develop a backtracking-based solution. Apply pruning techniques to discard invalid sets.
Justify how your solution works and explain NP-Complete classification
Your Code
Input
Output
Activate Windows
Go to Settings to activate Windows.
Submitted

Code of Conduct

I Tejaswi Reddy K (192371036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Marks Evaluation:

Question No.	Problem Understanding (25%)	Constraint Analysis (25%)	Solution Development (25%)	Application of Concepts (15%)	Justification (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____